

Cuda number of elements is larger than assigned threads

Asked 2 years, 9 months ago Modified 2 years, 9 months ago Viewed 548 times

I am new to CUDA programming. I am curious that what happens if the number of elements is larger than the number of threads?

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In this simple vector_add example

```
__global__
void add(int n, float *x, float *y)
{
    int i = blockIdx.x * blockDim.x + threadIdx.x;
    if (i < n)
        y[i] = x[i] + y[i];
}
```

Say the number of array elements is 10,000,000. And we call this function using 64 blocks and 256 threads per block:

```
int n = 1e8;
int grid_size = 64;
int block_sie = 256;
```

Then, only $64 \times 256 = 16384$ threads are assigned, what would happen to the rest of the array elements?

c++

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asked Nov 12, 2021 at 0:35

Yihan Hu

1137

1 Answer

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what would happen to the rest of the array elements?

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Nothing at all. They wouldn't be touched and would remain unchanged. Of course, your `x` array elements don't change anyway. So we are referring to `y` here. The values of `y[0..16383]` would reflect the result of the vector add. The values of `y[16384..9999999]` would be unchanged.

For this reason (to conveniently handle arbitrary data set sizes independent of the chosen grid size), people sometimes suggest a [grid-stride-loop](#) kernel design.

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answered Nov 12, 2021 at 0:42

Robert Crovella

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What will happen if we assign fewer number of threads in a vector of size N? Should the number of threads be the same as number of elements? – Mahdi Amrollahi

Dec 8, 2023 at 7:09